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Exploring Fusion Strategies for Multi-Modal Pneumonia Classification with Modality-Specific Explainability

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***Abstract*—Pneumonia diagnosis using chest radiographs has been widely studied thoroughly across several research, but most approaches rely solely on imaging data, often overlooking the valuable metadata associated with medical imaging.In this study, we explore the impact of multi-modal learning utilizing data fusion by combining chest X-ray images with tabular metadata (including age, gender, view position, and pixel spacing) for binary classification of Pneumonia vs. No Finding using a subset of the National Institutes of Health (NIH) Chest X-ray dataset. We analyzed three fusion strategies: early fusion, intermediate fusion, and late fusion, each integrating image and tabular modalities at different stages of the architecture. Furthermore, we also anaylzed the performance of using a single modality (image data only) to further justify the superiority of multi-modal tasks in the context of medical image classification. To ensure a fair comparison, all experiments are conducted on a balanced subset of 10,000 samples (5,000 per class) with consistent architectures. Our results show that intermediate fusion and late fusion achieves the best performance, outperforming both single modality and other fusion strategies in accuracy, F1-score, and AUC-ROC. Transparency of the model is further enhanced by incorporating GradCAM for visual explainability and LIME for tabular feature interpretation. This study highlights the complementary advantages of multi-modal fusion and provides insight into the role of metadata in medical image classification.**

***Index Terms*—Fusion, CXR Image, Classification, Multi-Modal, GradCAM, LIME, XAI, Deep Learning**

I. Introduction

Chest X-ray radiography (CXR) is the most commonly conducted medical imaging procedure and remains a fundamental first-line diagnostic tool for detecting a range of thoracic conditions, including pneumonia, pneumothorax, and lung nodules [1]. However, interpretation of CXRs is operator-dependent, time consuming and may lead to delays in diagnosis, especially in high-volume or resource-limited clinical settings. With around two billion radiographs performed globally every year, there is an urgent need for automated systems that can help radiologists in triaging abnormal studies and alleviate workload [2]. Deep learning, especially Convolutional Neural Networks (CNN)-based methods for chest X-ray interpretation have become prominent, successfully detecting multiple thoracic pathologies [1]. However, most existing approaches rely solely on imaging data. This unimodal focus neglects readily available patient metadata such as age, gender, view position, and acquisition parameters, which can provide complementary diagnostic context. Multi-modal research in medical imaging aggregates multiple types of data (imaging, clinical reports, table based metadata) to improve the diagnostic accuracy and the robustness of the model.

In addition, XAI has an important role in providing explainable insights into the model decision [3], addressing the challenge of deep learning being ”black box”. Explainable AI methods, including attention visualization and assessments of feature relevance, enable clinicians to better understand and trust the system's decisions, which supports the integration of automated solutions in critical healthcare settings. One possible first step, would be to determine normal and abnormal scans. An early dismissal of normals allows the clinician to focus on characterizing the precise nature of the abnormality in the remaining studies. We have incorporated these in our research for model generalizability and explainability.

Therefore our main contributions are:

- Proposed lightweight architecture incorporating fusion strategies by utilizing the MobileNetV3 model which is parameter efficient.
- Systematic comparison of the fusion Strategies along with the perfomance of Single Modality for pneumonia classification
- Modality-Specific Explainability, utilizing GradCAM and LIME for Image and tabular data respectively, to enhance transparency for multi-modal tasks.

II. Literature Review

Pneumonia detection on chest radiographs has been heavily investigated by using convolutional neural networks (CNNs) on datasets like NIH ChestX-ray14 [4]. Some other works have used CNNs, such as MobileNet, VGG for classification of brain tumors [5] or classic ML methods like hybrid linear regression to predict the risk of cardiovascular disease [6]. As the tasks become more challenging, more and more works

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TABLE I: Summary of Literature on Pneumonia Classification with Fusion and XAI

Reference	Modality Used	XAI Used	Limitations
[4]	CXR Only	No	Focused only on image classification, no fusion or XAI
[5]	MRI Only	No	No explainability
[6]	Tabular data	No	No fusion utilized
[7]	Image, Report and Demographic	Yes	XAI was not used for all modalities
[8]	Image and Audio	No	Explainability not provided
[9]	Images and Tabular	No	No explainability
[10]	Image and Tabular	Yes	Limited Samples
[11]	Image and Tabular	No	No Explainability
[12]	CXR Only	Yes	Fusion not utilized

explore multimodal approaches. For instance, [7] integrated patient demographics, reports and images with expert explainability; while the authors of [8] developed a classifier for respiratory diseases using lung sounds spectrograms and obtained an accuracy of over 95%. The authors in [9] concatenated images and tabular data from MIMIC-CXR, achieving an accuracy of 92.77%, with intermediate fusion achieving the highest performance. Similarly, [10] combined CT and U-Net with graphbased features and transformers for pneumonia screening, achieving a 94% accuracy with improved interpretability. Hsieh et al. [11] introduced MDFNet, a multimodal Mask R-CNN–based method integrating X-rays and clinical information, which achieved 12% better localization. Finally, [12] assessed interpretability techniques for pneumonia detection on ResNet50, where Layer-wise Relevance Propagation (LRP) proved the most effective method while obtaining a diagnostic accuracy of 0.91 with high interpretability (Mean Relevance Score)=0.85) suitable for clinical deployment.

A. Research Gap

In summary, while there exists a lot of research in image-based Pneumonia classification, gaps remain in three major areas: (1) systematic comparison of fusion strategies, (2) balanced evaluation with a clean set of data to avoid class imbalance bias, and (3) joint application of GradCAM and LIME to interpret model behavior across modalities. Our work addresses these research gaps by evaluating early, intermediate, and late fusion strategies using a balanced subset of the NIH dataset containing 5000 Pneumonia and 5000 No Finding samples. Furthermore, we utilized GradCAM and LIME to provide end-to-end explainability for both image and tabular data, contributing a novel benchmark for interpretable multimodal classification. Table I provides the entire summary of all the reviewed literature.

III. Dataset

The utilized dataset in this research is the medical imaging dataset from the (NIH) Clinical Center [3]. It is composed of chest X-ray in the frontal view with 112,120 images from 30,805 unique patients over 1992-2015, each annotated for the presence or absence of 14 common thoracic diseases

TABLE II: Tabular Features and Their Preprocessing Techniques

Feature Name	Original Data Type	Preprocessing Applied
Age	Numerical (Continuous)	Min-Max normalized to the [0, 1]
Gender	Categorical (Binary: M/F)	One-hot Encoding
View Position	Categorical (Binary: PA/AP)	One-hot Encoding
Pixel Spacing	Numerical (Continuous)	Median imputation

Tabular features of the NIH Chest X-ray Dataset (defined in Table II) undergo preprocessing with the following steps, to render them more suitable for machine learning. The continuous numeric value of ”age” is min-max normalized into the interval between 0 and 1 in order to standardize the scale. Both Gender and View Position, categorical binary features (M/F and PA/AP) are one-hot encoded to represent their inference in a numerical format to be used in the model. Another numeric feature: Pixel Spacing is preprocessed by median imputation for missing values.

Table III summarizes the distribution of No Finding and Pneumonia binary classification. The fair split, 8:1:1, allows for strong classifiers and fair evaluation, taking advantage of metadata of the dataset.

TABLE III: Dataset Split for Binary Classification of Chest X-rays (Pneumonia vs No Findings)

Class	Training Set	Validation Set	Test Set
Pneumonia	4000	500	500
No Findings	4000	500	500
Total Images	8000	1000	1000

IV. Methodology

We present a multimodal Pneumonia detection model, in which we have associated the chest X-Ray images with tabular data (age, gender, view position and pixel spacing).

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For tabular, we consider normalization, one-hot encoding and median imputation; for image, we use resampling, resizing and augmentation by rotation for robustness. We compare four fusion strategies, early, intermediate, late and a single-modality baseline (images only) using MobileNetV3 Small adapted to different settings. Early fusion fuses image and tabular features at the input layer, intermediate fusion integrates them at one hidden layer, and late fusion aggregates predictions of distinct image and tabular models. The unimodality model uses information of only images for comparison.

A. Evaluation

The performance was tested against F1-score recall and latency presented. For interpretability, we highlighted important image regions by Grad-CAM and explained predictions locally using LIME securing both global and local explanations. This approach improves trust and clinical relevance. Figure 1 show the entire workflow of our proposed strategy.

B. Experimental Setup

We experimented in Python 3.8 using PyTorch 2.x, NumPy, SciPy, and Scikit-learn. Training was performed on a workstation with an AMD Ryzen 5 5600G CPU, 8GB RAM, and an NVIDIA RTX 3060 GPU (8GB VRAM). We trained the models using the RMSProp optimizer and BCEWithLogitsLoss2. To optimize classification and account for class imbalance, a threshold of $\tau = 0.3819$ was applied to the sigmoid outputs. To ensure reproducibility, each configuration was repeated with 10 independent cycles with unique random seeds and dataset shuffling.

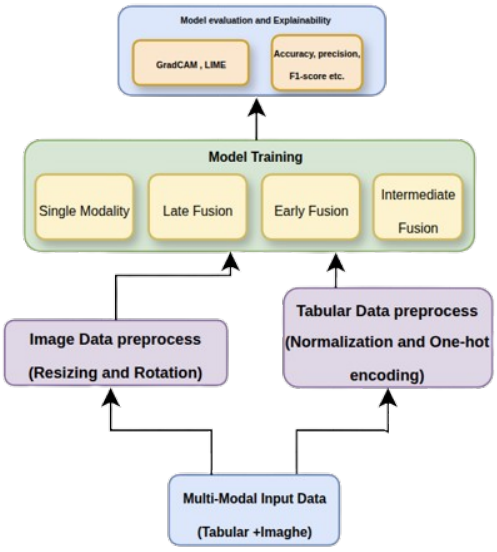


Fig. 1: Workflow

V. Fusion Models Used

To utilize both image-based and structured data, we explored different approaches for integrating chest radiographs with accompanying clinical attributes. These fusion methods were designed to enhance the quality of classification by enabling the model to draw insights from diverse modalities. Our work involved evaluating three types of multi-modal fusion—early, intermediate, and late—while also benchmarking against a single-modality baseline that relied solely on X-ray images. Each fusion strategy integrates the two modalities at different stages of the network and exhibits a unique trade-off between complexity and expressiveness. Figure 2 shows the three different fusion architectures used for comparison. All the fusion strategies incorporates a Neural Network classifier where the first layers maps the extracted features to a 256-dimensional embedding followed by a ReLu activation and a dropout with probability of 0.3. Finally this 256-dimensional embedding is mapped to the output layer with a single neuron where softmax is applied to evaluate the final probability.

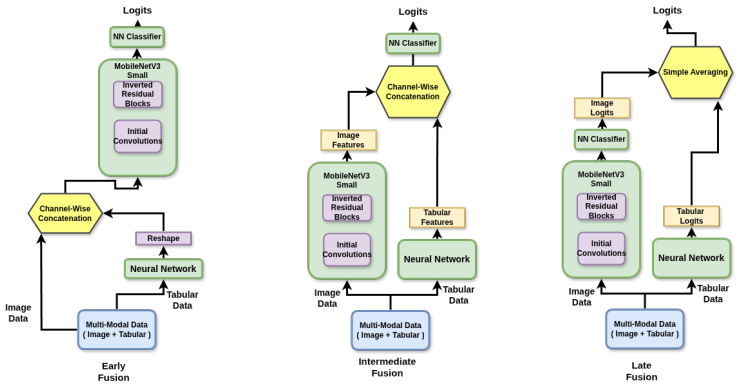


Fig. 2: Fusion Strategies (Early Fusion, Intermediate Fusion and Late Fusion)

A. Early Fusion

In this strategy, the image and tabular inputs are combined after basic preprocessing. In our implementation, the tabular features are passes through a neural network and reshaped to match the spatial dimensions of the image tensor, allowing the combined tensor to pass through a shared MobileNetV3 backbone. This approach enables the model to learn joint low-level representations across both modalities. While early fusion offers a uniform input structure, it can be limited due to differences in data modalities and scales. Equation 1 shows the early fusion operation where x_{img} is the image tensor, $\tilde{x}_{\text{tab}}$ if the tabular feature tensor, x_{fused} represents the combined tensor after concatenation, f_{backbone} is the model for extracting features and $\hat{y}$ is the output logits.

$$x_{\text{fused}} = \text{Concat}(x_{\text{img}}, \tilde{x}_{\text{tab}}), \quad \hat{y} = f_{\text{backbone}}(x_{\text{fused}}) \tag{1}$$

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B. Intermediate Fusion

Intermediate fusion occurs at the feature level, where each modality is first processed independently to extract high-level representations. In our model, features from the image are extracted using a MobileNetV3-Small backbone, while the tabular data is processed through a fully connected neural network. The feature outputs from different modalities are merged across channels and passed into a common classification network. By first extracting distinct representations from each data source and then integrating them, the model is able to capture complementary information. Our findings indicate that this method achieved the best results across all evaluation measures. Equation 2 shows the intermediate fusion operation where f_{img} is the image feature extractor (MobileNetV3) and f_{tab} if the tabular feature extractor (Neural Network).

$$z = \text{Concat}(f_{\text{img}}(x_{\text{img}}), f_{\text{tab}}(x_{\text{tab}})), \quad \hat{y} = f_{\text{cls}}(z) \tag{2}$$

C. Late Fusion

Late fusion involves training independent sub-networks for image and tabular data, each producing a separate output logits. The final prediction is obtained by averaging or weighting these outputs. In our case, the two logits are simply averaged. This method treats each modality as an expert, allowing independent feature extraction but limiting the opportunity for joint feature learning. It is relatively easier to implement and computationally efficient but may underperform when strong inter-modality interactions are needed. Equation 3 summarizes the late fusion operation.

$$\hat{y}_{\text{img}} = f_{\text{img}}(x_{\text{img}}), \quad \hat{y}_{\text{tab}} = f_{\text{tab}}(x_{\text{tab}}), \quad \hat{y} = \frac{1}{2}(\hat{y}_{\text{img}} + \hat{y}_{\text{tab}}) \tag{3}$$

D. Uni-modal (Image Only)

As a baseline, we trained a uni-modal model using only chest X-ray images with the MobileNetV3-Small architecture. This model excludes tabular data entirely, relying solely on image-based features for classification. It serves as a useful reference, for comparing with the fusion-based approaches, to demonstrate the gain achieved by fusion.

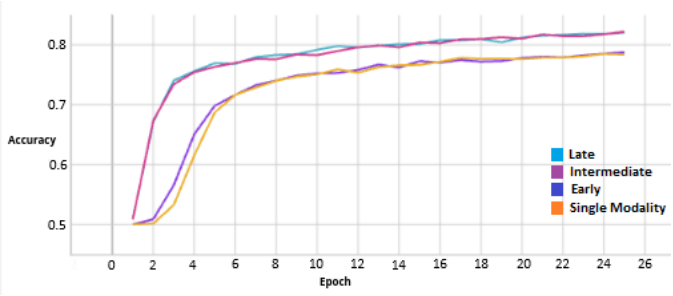


Fig. 3: Training accuracy across epochs for all compared fusion strategies, displaying performance dynamics between single modality, early, intermediate, and late fusion models

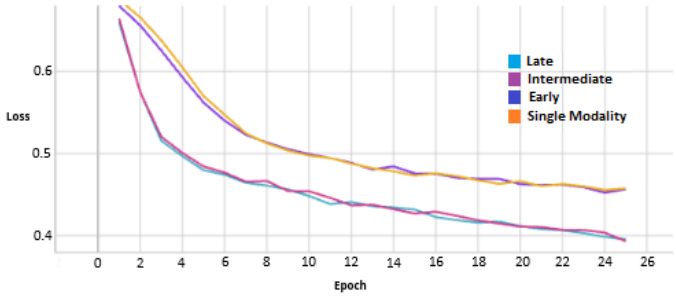


Fig. 4: Epoch-wise loss curves for single modality and multimodal fusion approaches, highlighting model convergence rates and stability during training

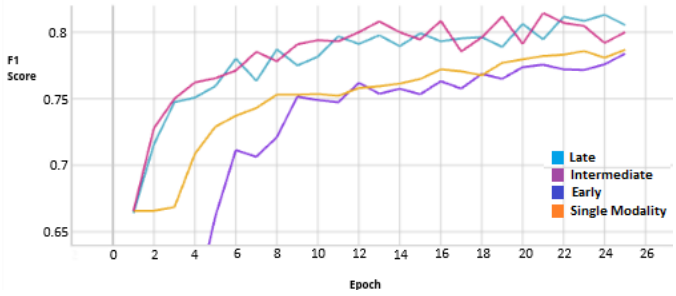


Fig. 5: Evolution of F1-score per epoch for different fusion methods, illustrating improvements in balanced classification as fusion depth increases.

VI. Result Analysis

We performed an extensive comparison of the proposed fusion strategies across multiple performance and efficiency metrics, as summarized in Table IV. In addition, every model underwent ten independent training and testing cycles, each initialized with a unique random seed to capture variability from random initialization and dataset ordering. The reported accuracy represents the average performance across these runs, where we used a 95% confidence interval threshold for the estimation using Student's t-distribution. The strategies compared include: Single Modality (image-only), Early Fusion, Intermediate Fusion, and Late Fusion.

Single Modality (image-only) serves as the baseline and achieved an accuracy of 76%, with relatively low computational cost (0.23 ms latency, 1.862 GFLOPs, 1.07 M parameters). This indicates that visual features alone carry moderately discriminative power, but does not include the contextual understanding that metadata can provide.

Early Fusion showed a slight improvement in performance (77% accuracy), by incorporating tabular features early in the network pipeline. The gain, though modest, comes without significant computational overhead (0.24 ms latency), confirming that metadata contributes marginally at the early stages of the architecture.

Intermediate Fusion outperformed all other strategies, achieving the highest accuracy of 80%, with an AUC-ROC of 0.8717 (Figure 7), and superior performance in both precision

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TABLE IV: Comparison of Fusion Strategies Based on Performance, Complexity, and Efficiency Metrics

Fusion Strategy	Accuracy	F1-Score	Precision	Recall	AUC-ROC	Latency	FLOPs	Params
Single Modality	0.76 ± 0.0036	0.77 ± 0.0029	0.75 ± 0.0018	0.78 ± 0.0033	0.83	0.23ms	1.862G	1.07M
Early Fusion	0.77 ± 0.0026	0.76 ± 0.0024	0.80 ± 0.0021	0.74 ± 0.0042	0.84	0.24ms	1.863G	1.09M
Intermediate Fusion	0.80 ± 0.0029	0.80 ± 0.0041	0.79 ± 0.0026	0.81 ± 0.0037	0.87	0.32ms	1.863G	1.09M
Late Fusion	0.79 ± 0.0051	0.80 ± 0.0017	0.81 ± 0.0043	0.80 ± 0.0025	0.85	0.24ms	1.863G	1.09M

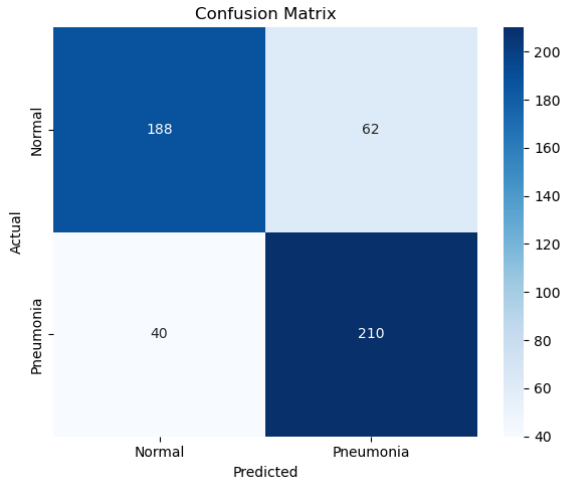


Fig. 6: Confusion Matrix on Test Set of Intermediate Fusion Model

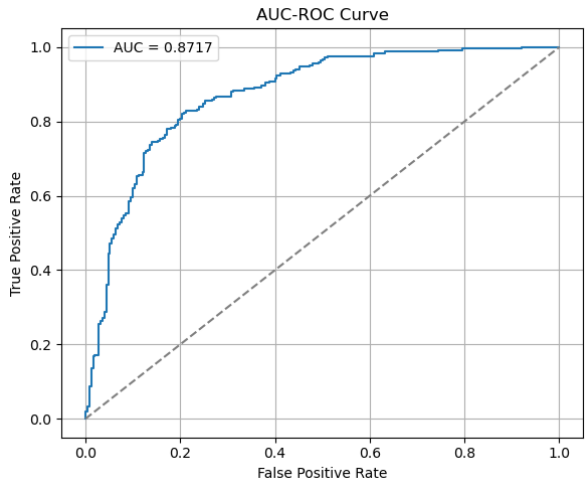


Fig. 7: ROC curve of Intermediate Fusion Model

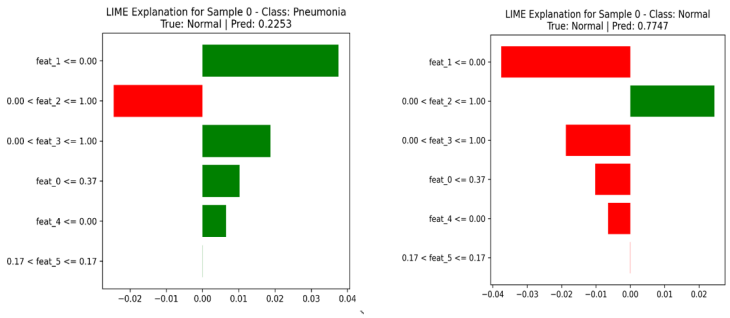


Fig. 8: LIME explanation for Normal and Pneumonia class. Features: (feat_1,2: gender; feat_3,4: view position; feat_0: age; feat_5: pixel spacing).

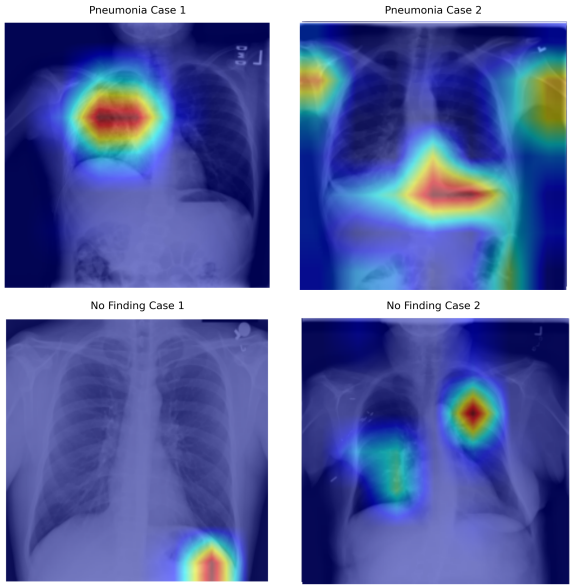


Fig. 9: GradCAM Images on Normal and Pneumonia Class

and recall. The confusion matrix (Figure 6) also shows better classification of both Pneumonia and No Finding classes. Although this model has slightly higher latency (0.32 ms), it benefits from combining image features with tabular representations at a semantically meaningful level. This suggests that modality interaction at the feature embedding level allows for better decision boundaries. The loss, accuracy and F1-score curves shown in figure 4, 3 and 5

Late Fusion also showed competitive results (79% accuracy), with performance approaching that of intermediate fusion. Since the two modalities are processed independently and combined at the decision level, this approach preserves computational efficiency (0.24 ms latency) while still capturing complementary signals. However, it may lose cross-modal interactions that intermediate fusion can leverage.

Overall, the results confirm that fusing metadata with image features enhances Pneumonia classification performance. Among all approaches, Intermediate Fusion emerges as the

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most effective strategy, offering the best trade-off between accuracy and computational cost. Additionally, we also carried out further experiements to test for generalizability of our approach by going with a multi-class classification that incorporates 3 different classes of Pneumonia, No Finding and Effusion. In this experiment, Intermediate Fusion again takes the lead along with Late Fusion and Early Fusion produced the least scores. The results are reported in Table V. This further proves that the findings are consistent at multi-class approaches as well. Moreover, its compatibility with modality-specific explainability such as GradCAM and LIME further supports its suitability for clinical applications.

TABLE V: Evaluation Metrics on Multi-Class Classification (Macro-Averaged)

Fusion Strategy	Accuracy	Precision	Recall	F1-score
Early Fusion	0.6011	0.5963	0.5370	0.5650
Intermediate Fusion	0.7389	0.7608	0.7110	0.7350
Late Fusion	0.7200	0.7190	0.7286	0.7237

VII. XAI

Explainable AI (XAI) explores the interpretability of the multi-modal pneumonia classification model using methods such as LIME and GradCAM to offer per modality insights. To interpret the influence of tabular features on the classification outcome, we employed LIME (Local Interpretable Model- agnostic Explanations). As shown in Figure 8, the most in- fluential feature for the "Normal" prediction was the patient's gender. Specifically, the presence of the one-hot encoded "Female" feature $(feat_2 = 1)$ contributed much positively, while the corresponding "Male" feature $(feat_1 = 0)$ had a good negative effect. This inverse relationship is expected due to one-hot encoding. The view position $(feat_3, representingPA)$ showed a moderate negative contribution, suggesting the model associated PA views with slightly higher pneumonia likelihood in this case. The patient's age $(feat \leq 0.37, age \leq 37)$ also partially decreased the model's confidence for the Normal class. Pixel spacing $(feat_5)$ had minimal influence in this prediction. Overall, the LIME output provided transparent insight into how each metadata feature contributed to the model's decision. Figure 9 shows the GradCAM visualizations of the Normal and Pneumonia classes, providing spatial explainability in terms of heatmaps superimposed on the X-ray images. For pneumonia-case-1, the heatmaps focus on lung regions, which are clinically reasonable. Also, for No Finding Case 1 images on the other hand exhibit very low levels of activations with only on regions outside the lung which highlights model's capability of correctly indicating lack of pneumonia. Conversely, the right-side image (2nd cases) shows a misclassification, where the model's attention and confidence differed, helping to identify potential reasons for the error. Together, these XAI methods provide both global and local transparency across modalities, fostering clinical acceptability of the model by elucidating how both imaging and metadata contribute to final decisions.

VIII. Conclusion and Future Work

This study compares early, intermediate, and late fusion approaches for multi-modal pneumonia classification on a balanced CXR dataset. Utilizing both image data and associated tabular metadata, our experiments demonstrated that intermediate fusion and late fusion consistently outperformed other strategies which is evaluated on various metrics. The incorporation of GradCAM and LIME enhanced model explainability by providing interpretable insights into image and tabular features. All these factors, prove the superiority of the multi-modal fusion in the context of pneumonia classification.

Despite fruitful results, several areas are still open for exploration. First, future work can extend this analysis to multi-class classification settings to distinguish between multiple thoracic diseases. Second, more advanced fusion mechanisms, such as attention-based fusion or Transformer-based multi-modal networks, can be analyzed to better capture inter-modality interactions. Third, the use of larger-scale datasets or real-world medical data may help validate and confirm the generalizability of the proposed approach. Lastly, enhancing explainability with newer techniques beyond GradCAM and LIME, such as SHAP or counterfactual explanations, also remains a valuable direction for contributions.

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